Journal of Inflammation Research

ISSN: 1178-7031 (Online) Journal homepage: www.tandfonline.com/journals/djir20

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To cite this article: Jianglong Chen, Yunfei Cui, Rujie Zhou, Yu Zhu, Meijia Chen, Jinhua Su, Jinhui Wang & Guang Li (2026) *Arundina graminifolia* Attenuates Renal Fibrosis in Chronic Kidney Disease by Modulating the TGF-β/PI3K-AKT/mTOR Pathway, Journal of Inflammation Research, , 579046, DOI: 10.2147/JIR.S579046

To link to this article: https://doi.org/10.2147/JIR.S579046

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Published online: 09 Jul 2026.

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Journal of Inflammation Research

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ORIGINAL RESEARCH

Arundina graminifolia Attenuates Renal Fibrosis in Chronic Kidney Disease by Modulating the TGF-β/PI3K-AKT/mTOR Pathway

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Objective: To explore the therapeutic effect and molecular mechanism of *Arundina graminifolia* (D.Don) Hochr (BYJ) on renal fibrosis in chronic kidney disease (CKD) mice.

Methods: Phytochemical profiles of the methanol extract BYJ were analyzed using ultra-performance liquid chromatography-quadrupole time-of-flight tandem mass spectrometry (UPLC-Q-TOF-MS), while network pharmacology and molecular docking were employed to predict the core active components and therapeutic targets. For experimental validation, a CKD mouse model was established via adenine induction. Mice were randomly assigned to normal control, model, and three BYJ treatment groups (low-dose: 2 g/kg·d, middle-dose: 4 g/kg·d, and high-dose: 8 g/kg·d, crude drug equivalent). Assessments included renal function parameters, histopathological changes, inflammatory markers, and oxidative stress indicators. The associated molecular mechanisms were analyzed using immunohistochemistry staining and Western blot.

Results: In silico analysis identified five core targets—PIK3R1, AKT1, SRC, MTOR, and EGFR—and suggested esculetin as a key active component with strong binding affinity to these targets. Experimentally, BYJ administration significantly reduced in serum creatinine (SCr), blood urea nitrogen (BUN), and the kidney index, and ameliorated glomerular atrophy and tubular dilation. Furthermore, BYJ treatment downregulated pro-inflammatory factors (eg, IL-1β), elevated superoxide dismutase (SOD) activity, and decreased malondialdehyde (MDA) content. The intervention suppressed the epithelial-mesenchymal transition (EMT) process, as evidenced by the upregulation of E-cadherin and downregulation of α-SMA expression. Mechanistic verification indicated that BYJ treatment was associated with attenuated the overactivated of TGF-β/PI3K-AKT/mTOR pathway in renal tissues.

Conclusion: The methanolic extract of BYJ effectively attenuates CKD progression in mice, which is associated with by modulated the overactivation of the TGF-β/PI3K-AKT/mTOR pathway, inhibited the EMT process, and ameliorated inflammatory response and oxidative stress. These findings provide a preliminary theoretical basis for the development of BYJ as a potential candidate agent for CKD treatment.

Keywords: *Arundina graminifolia*, chronic kidney disease, renal fibrosis, network pharmacology, TGF-β/PI3K-AKT/mTOR pathway, epithelial-mesenchymal transition

Introduction

Chronic kidney disease (CKD) is characterized by an irreversible reduction in renal function and structural degeneration of the kidneys, representing a prevalent degenerative condition of the urinary system globally. Globally, more than 10%

https://doi.org/10.2147/JIR.S579046
Received: 10 November 2025
Accepted: 18 May 2026
Published: 10 July 2026

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of adults are affected, with prevalence increasing annually.[1,2] In its first phases, CKD often manifests with nonspecific symptoms. It typically advances to end-stage kidney disease (ESKD), characterized by irreversible renal fibrosis. Dialysis or kidney transplantation is essential for patient survival at this stage, imposing a considerable financial burden on families and healthcare systems.[3]

The primary clinical characteristic of CKD progression is renal fibrosis, resulting from the interplay of various pathophysiological mechanisms, including the epithelial-mesenchymal transition (EMT).[4] EMT alters normal renal architecture by inducing epithelial cells to lose polarity and intercellular junctions, transdifferentiate into mesenchymal cells with migratory and fibrogenic properties, and promote excessive deposition of extracellular matrix (ECM), particularly type I collagen.[5] The dysregulated activation of the TGF-β/PI3K-AKT/mTOR signaling pathway is identified as a key molecular mechanism driving EMT. Renal fibrosis is expedited by the overactivation of this system, which inhibits epithelial indicators such as E-cadherin and elevates mesenchymal markers like α-SMA.[6–9] Renal damage may be aggravated by uncontrolled inflammation, marked by an overabundance of pro-inflammatory mediators such as IL-6, IL-1β, and TNF-α, as well as oxidative stress, shown by reduced superoxide dismutase (SOD) activity and elevated malondialdehyde (MDA) levels. In conjunction with EMT and fibrosis, these mechanisms establish a detrimental cycle that extends the duration of the disease.[10,11]

While angiotensin-converting enzyme (ACE) inhibitors, such as enalapril hydrochloride, are commonly used to alleviate renal burden, they rarely succeed in halting the EMT process or reversing pre-existing renal fibrosis. Furthermore, their long-term clinical application is often limited by adverse effects such as dry cough and hyperkalemia, creating a substantial unmet clinical need for effective anti-fibrotic therapies.[12–14] Notably, currently approved anti-fibrotic agents such as pirfenidone and nintedanib show only modest efficacy in renal fibrosis and are associated with significant gastrointestinal side effects, further highlighting the urgent need for novel therapeutic strategies. In recent years, natural bioactive compounds sourced from medicinal plants have emerged as promising therapeutic alternatives for the anti-fibrotic treatment of CKD. Unlike single-target synthetic drugs, multi-component herbal extracts possess the unique advantage of exerting synergistic effects at multiple biological nodes, making them particularly suitable for the complex, multi-factorial nature of CKD. However, this multi-component nature also presents significant challenges in identifying specific active constituents and their precise molecular mechanisms, which has hindered the clinical translation of many herbal medicines. Although the chemical complexity of these extracts presents challenges in identifying specific molecular mechanisms, the integration of systems-level approaches like network pharmacology enables systematic screening of core active constituents and their corresponding regulatory targets.[15–17] These findings highlight the importance of understanding the mechanisms by which plant-derived chemicals modulate key fibrotic signaling pathways and the EMT to prevent and treat CKD.

According to books such as *Dian Nan Ben Cao* and *Chinese Dai Medicine, Arundina graminifolia* (D. Don) Hochr (BYJ) is traditionally used for detoxification, diuresis, heat-clearing and urinary retention relief. It has been used to treat edema and anuria, clinical manifestations closely associated with renal failure. Phytochemical studies have shown that BYJ is abundant in flavonoids, steroidal saponins and alkaloids, which have anti-inflammatory, antioxidant and cell phenotype-modulating activities, which provides a pharmacological foundation for its potential renoprotective effects.[18–20] Our previous studies have shown that in a mouse model of cisplatin-induced renal injury, the methanolic extract of BYJ significantly reduced blood urea nitrogen (BUN) and serum creatinine (Cr) levels, attenuated renal interstitial collagen fiber deposition, and downregulated the expression of interstitial markers such as α-SMA. These findings indicate that BYJ may partially alleviate fibrosis by modulating the EMT process. However, these earlier findings were primarily limited to phenotypic and functional protection in acute renal injury. The specific mechanistic gap addressed in this study is how BYJ modulates the chronic, progressive fibrotic process—particularly the transition from persistent inflammation to EMT-mediated renal structural remodeling—rather than simply confirming its renoprotective phenotype.

While multiple pathways like Wnt/β-catenin contribute to renal damage, we prioritized the TGF-β/PI3K-AKT/mTOR signaling axis as the central focus of this study. This pathway serves as a critical integration hub where TGF-β triggers the downstream PI3K-AKT-mTOR signaling to coordinate protein synthesis, metabolic reprogramming, and the cytoskeletal rearrangements necessary for mesenchymal transdifferentiation. This research integrates phytochemical profiling,

network pharmacology, molecular docking, and in vivo validation to elucidate the EMT-targeted anti-fibrotic effects of BYJ, thereby providing a scientific basis for its development as a novel therapeutic agent for CKD.

Materials and Methods

Instruments

The analytical balance (DV215CD, precision 0.1 mg; Ohaus Corporation, USA), the ultrasonic cleaner (SB-800D; Ningbo Xinyi Ultrasonic Equipment Co., Ltd., China), the rotary evaporator (N-1300; Shanghai Ailang Instrument Co., Ltd., China), the vacuum drying oven (DZF-6092; Shanghai Yiheng Scientific Instrument Co., Ltd., China), the ultra-performance liquid chromatograph (LC400; Shimadzu Corporation, Japan), the gel imaging system (Tanon-5200Multi; Shanghai Tianneng Technology Co., Ltd., China), the optical microscope (DP53; Olympus Corporation, Japan), the tissue embedding system (Hisostar), the automatic tissue processor (MICROM STP120; Thermo Fisher Scientific, USA), automatic biochemical analyzer (SMT-120VP; Chengdu Smart Technology Co., Ltd., China), X500B Q-TOF time-of-flight mass spectrometer (AB SCIEX, Framingham, USA).

Drugs and Reagents

The medicinal materials were collected from the artificial cultivation base of the Yunnan Branch of the Institute of Medicinal Plant Development, Chinese Academy of Medical Sciences. They were authenticated as *Arundina graminifolia* (D. Don) Hochr. by Zhang Lixia, a researcher at the same institute (Herbarium Accession Number: 0052737, Figure S1). Methanol (Adamas, China), formic acid (HPLC grade, Aladdin, China), and acetonitrile (ACS/HPLC grade; Honeywell, USA) were utilized as solvents. Double-distilled water was supplied by Watsons Distilled Water Co., Ltd. (China). The following reagents were used: a seven-item renal function reagent disk (Chengdu Pulitai Biotechnology Co., Ltd., China); hematoxylin-eosin (HE) and Masson trichrome staining kits (Beijing Solarbio Science & Technology Co., Ltd., China); ELISA kits (Wuhan Servicebio Technology Co., Ltd., China); a bicinchoninic acid protein assay kit and an enhanced chemiluminescence detection kit (Kangwei Century Biotechnology Co., Ltd., China). Primary antibodies against PI3K, p-AKT, AKT, mTOR, E-cadherin, α-SMA, and TGF-β1 were purchased from Hua'an Biotechnology Co., Ltd. (China).

Animals

All C57BL/6J mice (4 weeks old, 22 ± 1 g) were purchased from Beijing Sibeifu Biotechnology Co., Ltd. and housed in specific pathogen-free (SPF) facility at the Yunnan Branch of the Institute of Medicinal Plants, Chinese Academy of Medical Sciences. Animals were maintained under 12-hour light/12-hour dark cycle with ad libitum access to standard chow and water. This study was approved by the Animal Ethics Committee of the Yunnan Branch of the Institute of Medicinal Plants, Chinese Academy of Medical Sciences (Approval No.: 2025039). All experimental procedures were performed in strict accordance with GB/T 42011–2022 *General Principles for the Welfare of Laboratory Animals*, the national standard for laboratory animal welfare in China, which aligns with the international guidelines of the World Organization for Animal Health (OIE) and covers the full lifecycle of laboratory animals including breeding, transportation, experimental use and humane disposal.

Preparation of the Administration Solution

Utilizing an ultrasonic cleaner, 100.00 g of BYJ coarse powder, which had been sieved over a 60-mesh screen, was removed on two occasions.[21] For each extraction, 1 L of chromatographic-grade methanol was introduced, followed by ultrasonication for 1 hour under the settings of 800 W power, 40 kHz frequency, and a temperature of 25 ± 2 °C. The two extracts were combined, filtered, and subjected to freeze-drying, initially pre-frozen for 24 hours at −80 °C, followed by 48 hours of freeze-drying under a vacuum pressure below 10 Pa. The extraction yield was 9.8% ± 0.3% (n = 3, dry weight). All dosages in this study are expressed as crude drug equivalent (CDE), meaning that 1 g/kg of BYJ treatment corresponds to 0.098 g/kg of dried methanol extract. The dried extract was stored at 4 °C until required again.

Mass Spectrometry Analysis

To prepare the test solution, freeze-dried BYJ powder (5.0000 ± 0.0005 mg, n = 3) was accurately weighed into a 50 mL amber-glass stoppered conical flask, dissolved in chromatographic-grade methanol, subjected to sonication (40 kHz, 15 min), and filtered through a 0.22 μm membrane.

Acquity HSS T3 column (2.1×100 mm, 1.8 μm) utilized in ultra-performance liquid chromatography-quadrupole time-of-flight tandem mass spectrometry (UPLC-Q-TOF-MS) (IDA mode). The mobile phase consists of 0.1% formic acid in water (A) or acetonitrile (B). Gradient: 0–2 min 5% B, 2–7 min 30% B, 7–10 min 30% to 60% B, 10–13 min 60% to 80% B, 13–17 min 90% B, 17–20 min 100% B, and 20–23 min 5% B. Column temperature: 40°C; injection volume: 5 μL; flow rate: 0.3 mL/min.

Mass spectrometry (X500BQ-TOF, ESI± modes): secondary scan range of 50–1200 Da, primary scan range of 100–1200 Da. N_2 utilized as a collision/nebulizing gas. Ionization voltage: ±5500 V; collision energy: ±10 V; orifice voltage: ±100 V; curtain gas pressure: 35 psi; nebulizer and auxiliary gas pressure: 60 psi each.

Animal Grouping and Treatment

No animals were excluded from the study, and all data points were included in the final statistical analysis. Thirty male C57BL/6J mice were randomly assigned to five groups (n = 6 per group) using a random number table: Control group: standard chow + oral gavage of 10 mL/kg distilled water; Adenine model group: 0.2% adenine-containing chow + oral gavage of 10 mL/kg distilled water; BYJ low-dose group: 0.2% adenine-containing chow + oral gavage of 2 g/kg BYJ (CDE, equivalent to 0.196 g/kg·d dried extract); BYJ middle-dose group: 0.2% adenine-containing chow + oral gavage of 4 g/kg BYJ (CDE, equivalent to 0.392 g/kg·d dried extract); BYJ high-dose group: 0.2% adenine-containing chow + oral gavage of 8 g/kg BYJ (CDE, equivalent to 0.784 g/kg·d dried extract). All treatments were administered once daily for four consecutive weeks.[22] All investigators performing histological evaluation and biochemical analyses were blinded to group assignments throughout the experiment.

Sample Collection and Processing

Mice were anesthetized with 1.5% isoflurane following the final administration and subjected to a 12-hour fasting period with free access to water. Prior to sample collection, anesthesia was maintained at 1.5% isoflurane to ensure loss of pain reflexes and physiological stability. After blood sampling, anesthesia was deepened to 3–4% isoflurane, and the animals were subsequently euthanized by cervical dislocation in accordance with the American Veterinary Medical Association (AVMA) Guidelines for the Euthanasia of Animals (2020). Orbital blood was allowed to clot for 1 hour at room temperature before centrifugation, and the serum was aliquoted and stored at −80°C for biochemical analysis. The right kidney was snap-frozen in liquid nitrogen and stored at −80°C for molecular analysis, while the left kidney was fixed in 4% paraformaldehyde (pH 7.4) for 24 hours for histological examination.

Pharmacological Network Analysis

Target Prediction and Screening

SwissADME was utilized to identify the active compounds in the BYJ extract, selecting those that adhered to Lipinski's Rule of Five and exhibited high gastrointestinal absorption. SwissTargetPrediction (http://www.swisstargetprediction.ch/) supplied the anticipated targets for each active chemical. CKD-related targets were retrieved from GeneCards (https://www.genecards.org) using the phrase "chronic kidney disease". A Venn diagram generated by bioinformatics tools (https://www.bioinformatics.com.cn/) was utilized to find overlapping drug-disease targets.

Network Visualization and Core Target Identification

An active component-intersection target network was constructed to identify significant active chemicals. The core targets were screened, and the overlapping targets were imported into the STRING database (https://cn.string-db.org/) to construct a protein-protein interaction (PPI) network. All network visualizations were generated using Cytoscape v3.7.0.

Pathway Enrichment Analysis

The Metascape database (http://metascape.org/) facilitated GO and KEGG enrichment studies, while bioinformatics tools were employed to illustrate the results.

Molecular Docking

The subsequent AlphaFold IDs were employed to forecast the three-dimensional structures of core targets: AF-P27986-F1-v4 (PIK3R1), AF-Q38998-F1-v4 (AKT1), AF-Q1JPZ3-F1-v4 (SRC), AF-P42345-F1-v4 (MTOR), and AF-P0CY46-F1-v4 (EGFR). AutoDock was utilized to execute molecular docking between significant drugs and primary targets, and binding energies were calculated. PyMOL version 2.1.0 was employed to visualize docking interactions.

Measurement of Serum Biochemical Parameters

The manufacturer's guidelines for the reagent kits were adhered to in order to ascertain the serum creatinine and blood urea nitrogen levels.

Renal Histopathology

Kidney tissues preserved in formalin were sectioned to a thickness of 5 μm, embedded in paraffin, and dehydrated with graded ethanol. For histological examination, sections were deparaffinized, rehydrated, and stained with Masson's trichrome and hematoxylin-eosin (HE).

Assessment of Lipid Peroxidation and Inflammatory Cytokines

Biochemical assays were conducted utilizing frozen kidney tissue and serum specimens preserved at −80°C. In accordance with the manufacturer's guidelines, commercial assay kits were employed to quantify the levels of SOD and MDA in renal tissue, together with the serum amounts of IL-1β, IL-6, and TNF-α. Each sample was analyzed in triplicate.

Immunohistochemistry (IHC)

Kidney tissue sections were dewaxed, rehydrated, and subsequently heated in 1× sodium citrate buffer for 10 minutes to facilitate antigen retrieval. A 10-minute treatment with 3% hydrogen peroxide, following chilling, inhibited endogenous peroxidase. Incubate for one hour at room temperature with 10% goat serum. Incubate primary antibodies (α-SMA 1:5000, PI3K 1:500, phosphorylated AKT 1:1000, mTOR 1:200, and E-cadherin 1:200) at 4 °C overnight. After three washes with PBS, incubate for one hour at room temperature with a species-specific secondary antibody prior to DAB staining. Nuclear staining with hematoxylin, followed by dehydration, mounting, and microscopic examination.

Western Blot

The kidney tissue was homogenized in RIPA lysis buffer containing 1% dual inhibitor, incubated on ice for two hours, and then centrifuged for 20 minutes at 12,000 rpm and 4°C. The BCA technique was employed to quantify the protein concentration. Ten percent SDS-PAGE was employed to separate 30 μg of protein per lane, thereafter transferred to PVDF membranes. Incubated for two hours at room temperature in TBST containing 5% skim milk for blocking. Primary antibodies (TGF-β1 1:2000, PI3K 1:2000, AKT and phosphorylated AKT 1:5000, mTOR 1:2000, E-cadherin, and α-smooth muscle actin 1:5000) were incubated overnight at 4°C. Subsequent to washing with TBST, incubate for two hours at ambient temperature with the secondary antibody. Image-Pro Plus 5.0 and TANON were employed for quantification, and ECL was utilized for development.

Statistical Analysis

Data were expressed as mean ± standard deviation (SD). Statistical comparisons were conducted using Student's *t*-test (two groups) or one-way analysis of variance (ANOVA) followed by Tukey's post-hoc test (multiple groups) in GraphPad Prism 8.0.2. Differences with $P < 0.05$ were considered statistically significant.

Results

Chemical Composition of BYJ Extract

Utilizing multi-stage mass spectrometric data, UPLC-Q-TOF-MS analysis, database inquiries, and literature citations, 25 chemical constituents, predominantly flavonoids and alkaloids, were identified in the methanol extract of BYJ. Table 1 includes detailed chemical profiles, whereas Figure 1 presents base peak chromatograms in both positive and negative ion modes.

Network Pharmacology Analysis

Active Component and Target Identification

A total of 13 active components and 624 potential targets were identified in BYJ. Additionally, the GeneCards database produced 5501 unique targets related with CKD. Intersection analysis revealed 456 overlapping targets between the active components and CKD (Figure 2A).

Active Component-Target Network Analysis

The five primary active constituents were identified using network topology analysis utilizing degree centrality: feruloyltyramine, diosmetin III, blumenol B 9-O-glucoside, coniferaldehyde, and esculetin (Figure 2B).

PPI Network Analysis

The top five important targets (PIK3R1, AKT1, SRC, MTOR, and EGFR) were identified based on composite scores that combine degree, betweenness centrality, and closeness centrality, after the construction of the PPI network using the STRING database and its visualization in Cytoscape (Figure 2C).

Table 1 Characterization of BYJ Component

No.	Name	Rt (min)	Precursor m/z	Adduct	Formula	Error/ppm
1	Elaidic acid	18.201	283.26298	$[M+H]^+$	$C_{18}H_{34}O_2$	−0.60
2	Linoleic acid	13.289	281.24713	$[M+H]^+$	$C_{18}H_{32}O_2$	−1.31
3	Iridin	12.503	545.12090	$[M+Na]^+$	$C_{24}H_{26}O_{13}$	1.65
4*	Aurantiamide acetate	12.056	445.21164	$[M+H]^+$	$C_{27}H_{28}N_2O_4$	−1.23
5*	Feruloyltyramine	8.237	314.13914	$[M+H]^+$	$C_{18}H_{19}NO_4$	1.49
6*	Coniferaldehyde	7.771	179.07034	$[M+H]^+$	$C_{10}H_{10}O_3$	0.39
7*	Loliolide	7.485	197.11711	$[M+H]^+$	$C_{11}H_{16}O_3$	−0.55
8	Nepetin-7-glucoside	7.340	501.10092	$[M+Na]^+$	$C_{22}H_{22}O_{12}$	−0.17
9*	Kaempferol	7.041	287.05489	$[M+H]^+$	$C_{15}H_{10}O_6$	−0.45
10*	Quercetin	6.847	303.04950	$[M+H]^+$	$C_{15}H_{10}O_7$	−1.41
11	Isoquercetin	6.847	465.10370	$[M+H]^+$	$C_{21}H_{20}O_{12}$	2.04
12	Saponaretin	6.742	433.11357	$[M+H]^+$	$C_{21}H_{20}O_{10}$	1.50
13	Hyperoside	6.653	465.10419	$[M+H]^+$	$C_{21}H_{20}O_{12}$	3.09
14	Rutin	6.653	611.16139	$[M+H]^+$	$C_{27}H_{30}O_{16}$	1.22
15*	Blumenol B 9-O-glucoside	5.971	411.19859	$[M+Na]^+$	$C_{19}H_{32}O_8$	−0.85
16	Hydrangeifolin I	5.934	439.15791	$[M+Na]^+$	$C_{19}H_{28}O_{10}$	1.00
17*	Phenylalanine	1.202	166.08675	$[M+H]^+$	$C_9H_{11}NO_2$	2.95
18*	Pyroglutamic Acid	1.166	130.04983	$[M+H]^+$	$C_5H_7NO_3$	−0.30
19	Stachydrine	0.862	144.10196	$[M+H]^+$	$C_7H_{13}NO_2$	0.34
20*	Trigonelline	0.844	138.05505	$[M+H]^+$	$C_7H_7NO_2$	0.65
21	Betaine	0.826	118.08595	$[M+H]^+$	$C_5H_{11}NO_2$	−2.62
22*	Pyrocatechol	1.189	109.02956	$[M-H]^-$	$C_6H_6O_2$	0.55
23	Orientin	6.417	447.09326	$[M-H]^-$	$C21H_{20}O_{11}$	−0.06
24*	Esculetin	5.674	177.01923	$[M-H]^-$	$C_9H_6O_4$	−0.56
25*	Batatasin III	10.436	243.10200	$[M-H]^-$	$C_{15}H_{16}O_3$	−2.71

Note: *indicates active ingredients of BYJ.

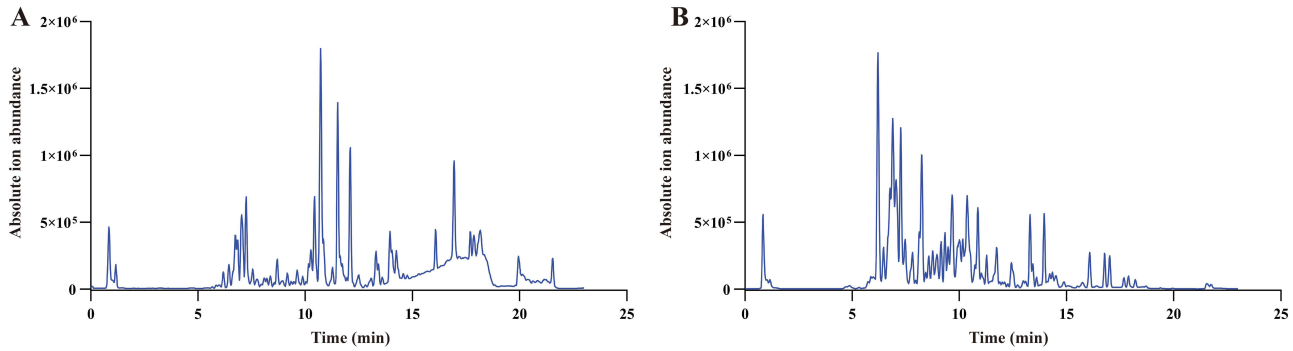


Figure 1 BPCs of BYJ methanol extract analyzed using ultra-high performance liquid chromatography-quadrupole time-of-flight tandem mass spectrometry. (**A**) BPC in positive ion mode. (**B**) BPC in negative ion mode.
Abbreviation: BPC, base peak chromatogram.

Pathway Enrichment Analysis

The PI3K-AKT, mTOR, and TGF-β signaling pathways exhibited the greatest enrichment among the 243 substantially enriched signaling pathways ($P < 0.05$) (Figure 2D). Furthermore, GO enrichment analysis indicated that BYJ intervention significantly influenced 454 molecular functions, 208 cellular components, and 2922 biological processes (Figure 2E).

Molecular Docking

The molecular docking results indicated that the esculetin, a key active component of BYJ, exhibited binding energies below −7.0 kcal/mol against all identified core targets. Compared with other major active constituents of BYJ, it demonstrated markedly stronger binding affinity to the core targets implicated in renal fibrosis (Figure 3A). The molecular interactions between esculetin and the selected core targets was visualized and analyzed by PyMOL (Figure 3B–F). These findings suggest that esculetin may serve as a potential key active component contributing to the anti-renal fibrosis effect of BYJ, and provides a clear direction for the subsequent isolation, purification of esculetin and validation experiments using the pure compound.

Effect of BYJ on Kidney Index, Morphology, and Renal Function Parameters

The kidneys of the model group had a solid texture, numerous dense white granules on the surface, and loose adhesion between the capsule and renal tissue, resulting in a paler appearance compared with the controls. Conversely, these degenerative changes exhibited varying degrees of improvement in mice administered BYJ (Figure 4A). As depicted in Figure 4B–D, the kidney index, serum creatinine, and blood urea nitrogen levels in the model group were significantly elevated compared to the control group ($P < 0.05$), increasing from 0.59 ± 0.098%, 18.68 ± 2.14 μmol/L, and 8.75 ± 1.03 mmol/L to 1.08 ± 0.16%, 39.28 ± 11.4 μmol/L, and 15.46 ± 2.49 mmol/L, respectively. BYJ therapy considerably reversed these increases ($P < 0.05$). The kidney index exhibited the most significant reduction in the high-dose BYJ group (0.71 ± 0.17%), whereas serum creatinine and blood urea nitrogen showed the greatest improvement in the middle-dose BYJ group (19.18 ± 2.15 μmol/L and 10.87 ± 0.90 mmol/L, respectively). These findings demonstrate that BYJ exerts a substantial therapeutic effect on adenine-induced renal injury, significantly diminishing elevations in serum creatinine, blood urea nitrogen, and kidney index.

Histopathological Analysis of Mouse Kidneys

Figure 5A illustrates kidney slices stained with hematoxylin and eosin (HE). The model group mice exhibited tubular dilatation, glomerular atrophy, and infiltration of inflammatory cells. However, a comparison between BYJ-treated mice and the model group demonstrated varying degrees of amelioration in these pathological alterations. The Masson staining results (Figure 5B and C) indicated that the model group exhibited considerably more renal fibrosis compared with the controls ($P < 0.05$), with the fibrotic area escalating from 4.61 ± 1.95% to 22.64 ± 4.25%. The BYJ treatment markedly reduced fibrosis ($P < 0.05$), with the middle-dose BYJ group demonstrating the most pronounced impact, decreasing the

https://doi.org/10.2147/JIR.S579046

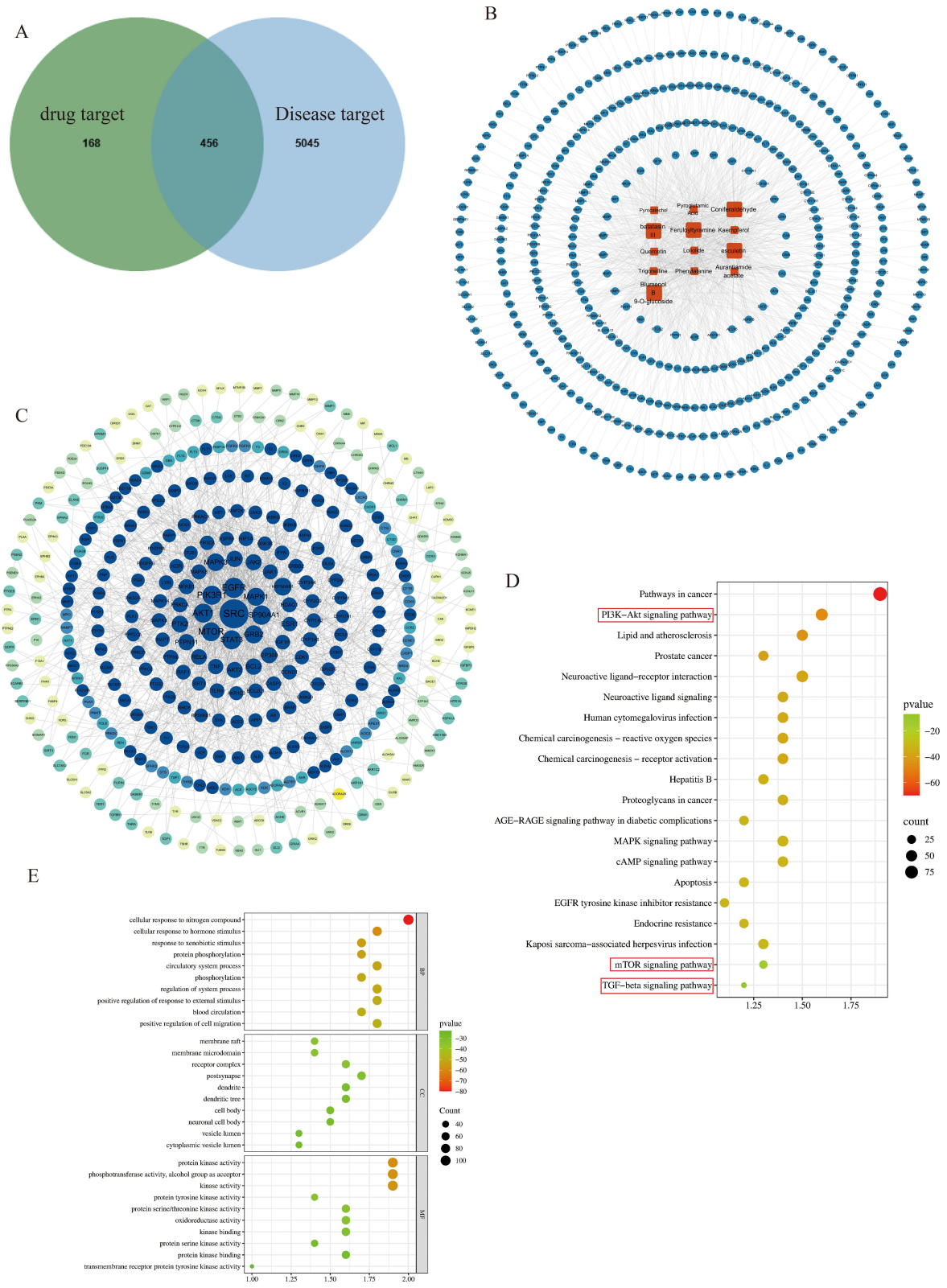


Figure 2 Network pharmacology analysis of BYJ in CKD. (**A**) Venn diagram showing the intersection of potential targets between AG active ingredient targets and CKD disease targets. (**B**) Network diagram of the "active ingredient-target" interactions. (**C**) Protein-protein interaction network of the 456 intersection targets; red nodes represent core targets. (**D**) Bubble plot of KEGG pathway enrichment analysis displaying core pathways. (**E**) Bar chart of GO enrichment analysis covering BPs, CCs and MFs.

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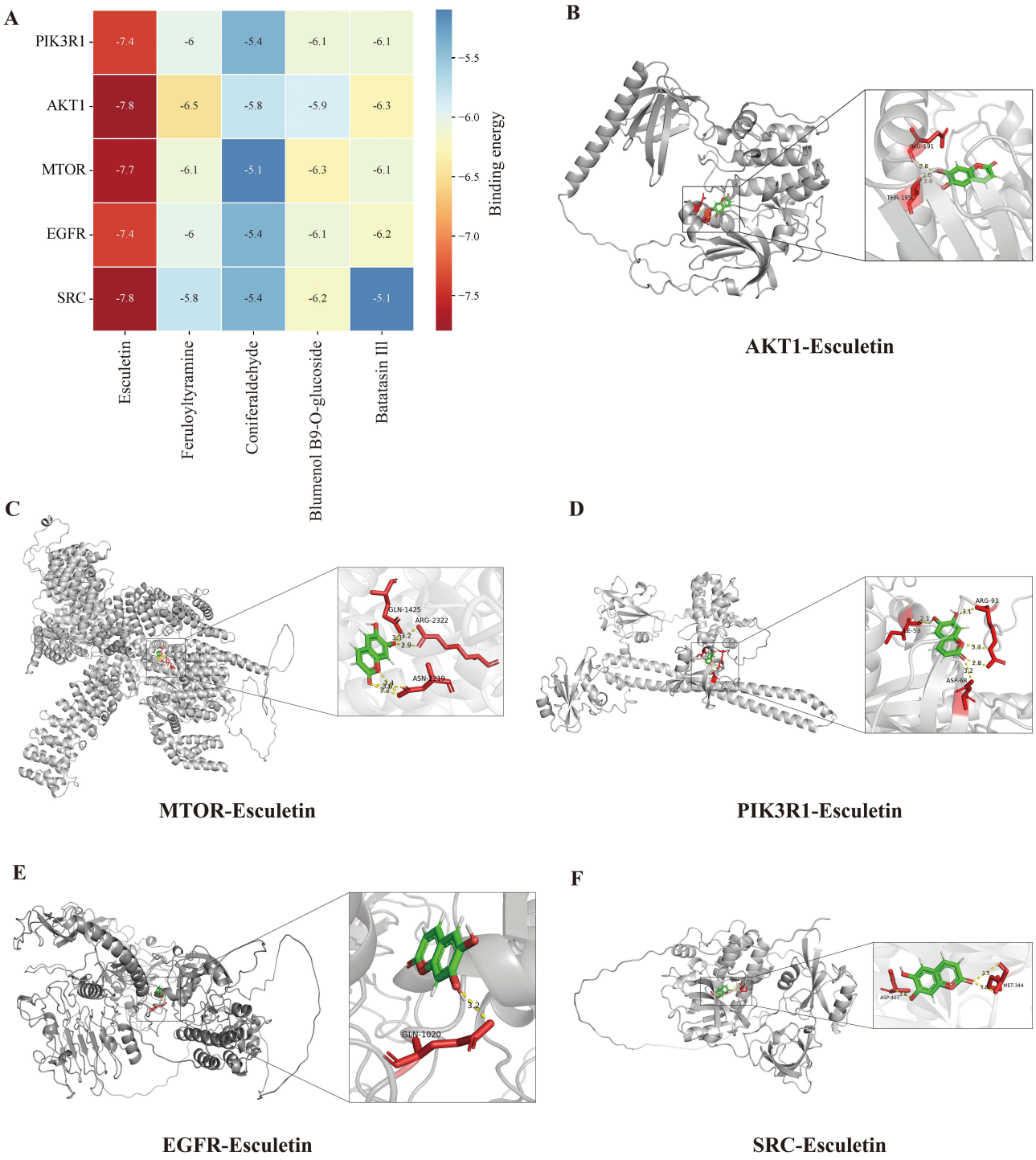


Figure 3 Molecular docking Molecular Docking of BYJ Active Components with Key CKD Targets. (**A**) Validation of molecular docking between core components and key targets. (**A**) Heatmap analysis of binding energies between core components and key targets. (**B**) Molecular docking pose of esculetin with the AKT1 target. (**C**) Molecular docking pose of esculetin with the MTOR target. (**D**) Molecular docking pose of esculetin with the PIK3R1 target. (**E**) Molecular docking pose of esculetin with the EGFR target. (**F**) Molecular docking pose of esculetin with the SRC target.

Abbreviation: PIK3R1, phosphatidylinositol 3-kinase regulatory subunit 1.

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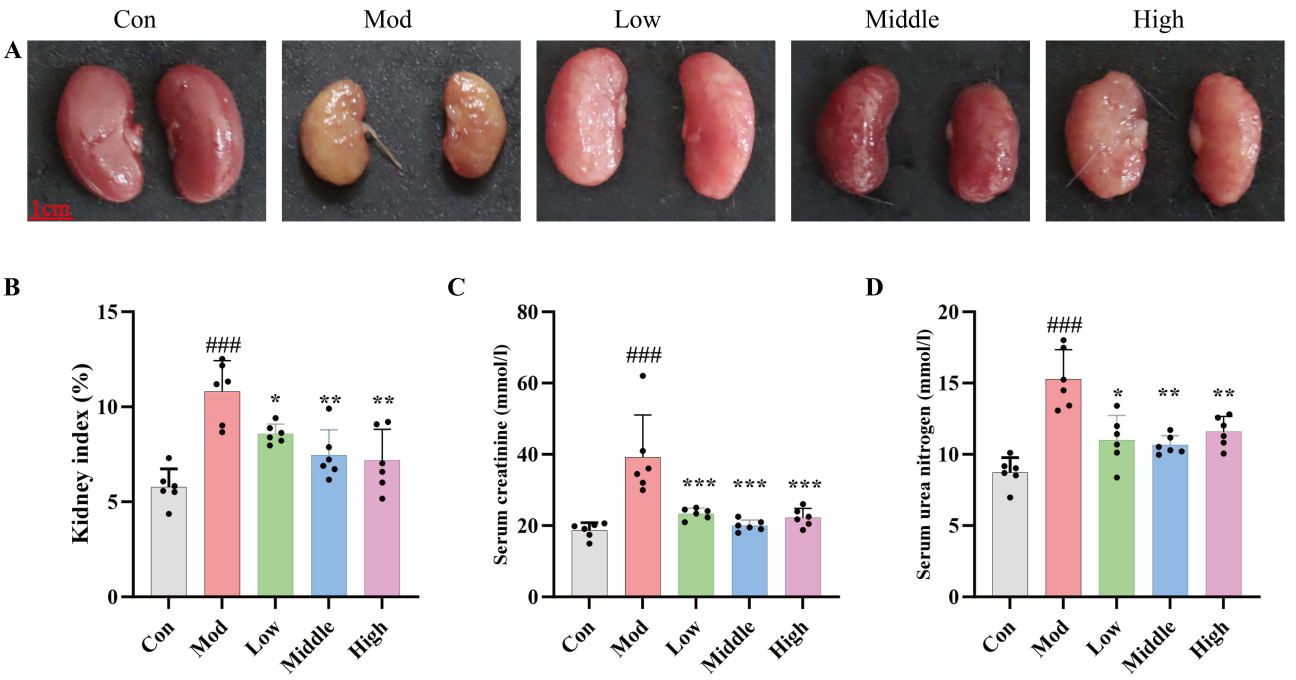


Figure 4 Results of morphological and biochemical indicators of mouse serum. (**A**) Macroscopic observation of mouse kidneys across all groups. Scale bar, 1 cm. (**B**) Statistical analysis of the kidney index. (**C**) Serum creatinine levels in each group. (**D**) Serum urea nitrogen levels in each group. ####P < 0.0001, compared with the Con group; *P < 0.05, **P < 0.01, ***P < 0.001, compared with the Mod group.

fibrotic area to 6.64 ± 0.94%. The results indicate that BYJ effectively mitigates renal damage and fibrosis induced by adenine.

Effects of BYJ on Inflammatory Cytokines and Lipid Peroxidation

Figure 6A–E demonstrates the impact of BYJ on inflammatory cytokines and lipid peroxidation; the model group exhibited a substantial dysregulation of these parameters ($P < 0.05$) relative to the control group. The levels of IL-1β, IL-6, and TNF-α increased from 36.23 ± 2.11 pg/mL, 37.75 ± 1.59 pg/mL, and 21.55 ± 1.50 pg/mL to 60.23 ± 3.82 pg/mL, 45.13 ± 1.61 pg/mL, and 42.07 ± 0.77 pg/mL, respectively. SOD diminished from 101.80 ± 5.32 U/g to 48.23 ± 5.82 U/g, whilst MDA escalated from 51.57 ± 1.39 mmol/g to 85.12 ± 2.88 mmol/g. BYJ therapy effectively restored these alterations ($P < 0.05$). IL-1β, TNF-α, MDA, and SOD exhibited the greatest enhancement in the high-dose BYJ cohort, adjusting to 46.87 ± 2.72 pg/mL, 34.18 ± 1.30 pg/mL, 68.12 ± 1.94 mmol/g, and 75.62 ± 2.86 U/g, respectively. The BYJ group with a middle dose exhibited the most significant effect on IL-6, reducing its concentration to 40.22 ± 1.56 pg/mL. These data indicate that BYJ mitigates oxidative damage and inflammation induced by adenine.

Inhibition of EMT in Renal Tissue

The expression of the epithelial marker E-cadherin in the renal tissue of the model group was considerably reduced ($P < 0.05$, from 1.00 ± 0.14 to 0.28 ± 0.02) compared to the control group, as shown by Western blot and IHC studies of α-SMA and E-cadherin (Figure 7A–E). The reduction was notably reversed ($P < 0.05$) in all BYJ-treated cohorts, with the highest-dose group exhibiting the most substantial effect (restoring E-cadherin to 0.88 ± 0.02). In contrast, the model group's expression of the mesenchymal marker α-SMA exhibited a significant increase ($P < 0.05$, from 1.00 ± 0.10 to 4.15 ± 0.15), while BYJ treatment markedly suppressed this expression ($P < 0.05$); the high-dose group demonstrated the most substantial inhibition, reducing α-SMA to 1.69 ± 0.16. These results indicate that BYJ mitigates renal fibrosis partially via upregulating E-cadherin and downregulating α-SMA, suggesting the inhibiting of EMT in the renal tissue.

Regulation of the TGF-β/PI3K-AKT/mTOR Signaling Pathway

The renal expression levels of TGF-β, PI3K, p-AKT, and mTOR were markedly elevated ($P < 0.05$) in the model group relative to the control group, as demonstrated by Western blot and IHC studies of pivotal proteins in the TGF-β/PI3K-

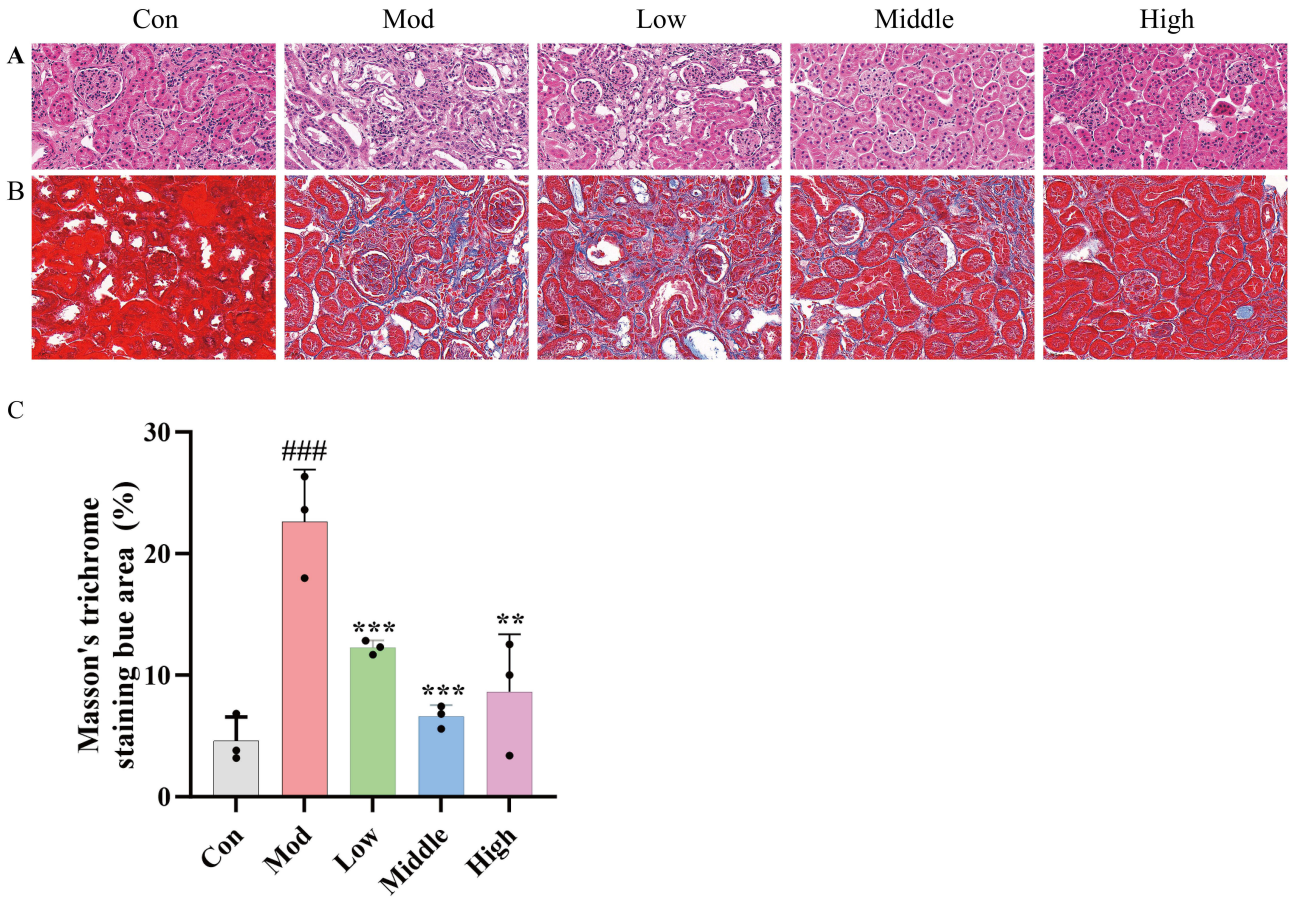


Figure 5 HE staining and Masson staining. (**A**) Representative images of H&E staining of kidney tissues (magnification, x200). (**B**) Masson's trichrome staining of kidney tissues illustrating collagen deposition (magnification, x200). (**C**) Statistical analysis of fibrotic area. ####P < 0.0001, compared with the Con group; **P < 0.01, ***P < 0.001, compared with the Mod group.

AKT/mTOR signaling pathway (Figure 8A–H). TGF-β rose from 1 ± 0.52 to 3.24 ± 0.14, PI3K from 1 ± 0.37 to 3.38 ± 0.21, p-AKT from 1 ± 0.33 to 3.40 ± 0.15, and mTOR from 1.00 ± 0.04 to 4.82 ± 0.22. This indicates that in adenine-induced kidney damage, the TGF-β/PI3K-AKT/mTOR signaling pathway is significantly overactivated, potentially contributing to the progression of EMT. The high-dose cohort exhibited the most pronounced effect, reducing TGF-β to 1.56 ± 0.09, PI3K to 1.61 ± 0.12, p-AKT to 1.84 ± 0.21, and mTOR to 1.81 ± 0.16. BYJ treatment markedly reduced these alterations ($P < 0.05$). These findings indicate that BYJ mitigates renal fibrosis by inhibiting the excessive activation of the TGF-β/PI3K-AKT/mTOR pathway, hence suppressing EMT.

Discussion

In this study, the dosage of BYJ for administered to mice was determined by converting the clinical dosage based on body surface area. The clinical daily dosage for adults (with a body weight of 70 kg) is 10–30 g/d.[23] Using the standard body surface area conversion factor of 9.1, the equivalent mouse dosage was calculated to be 1.29–3.89 g/kg/d. Additionally, based on the results of our research group's previous acute toxicity experiments and the study on the protective effect of BYJ against cisplatin-induced acute kidney injury,[21] 8 g/kg/d (calculated based on the crude drug amount) of BYJ exhibited no overt toxicity in mice while conferring renal protective effects. Based on the above equivalent dosage conversion results and the previous experimental basis, three dosage groups of BYJ, namely low, middle, and high, were set up in the chronic kidney injury model, with the dosages being 2, 4, and 8 g/kg/d, respectively.

UPLC-Q-TOF-MS analysis identified twenty-five chemical compounds in the BYJ methanol extract. Network pharmacology screening identified thirteen active compounds with favorable gastrointestinal absorption that comply

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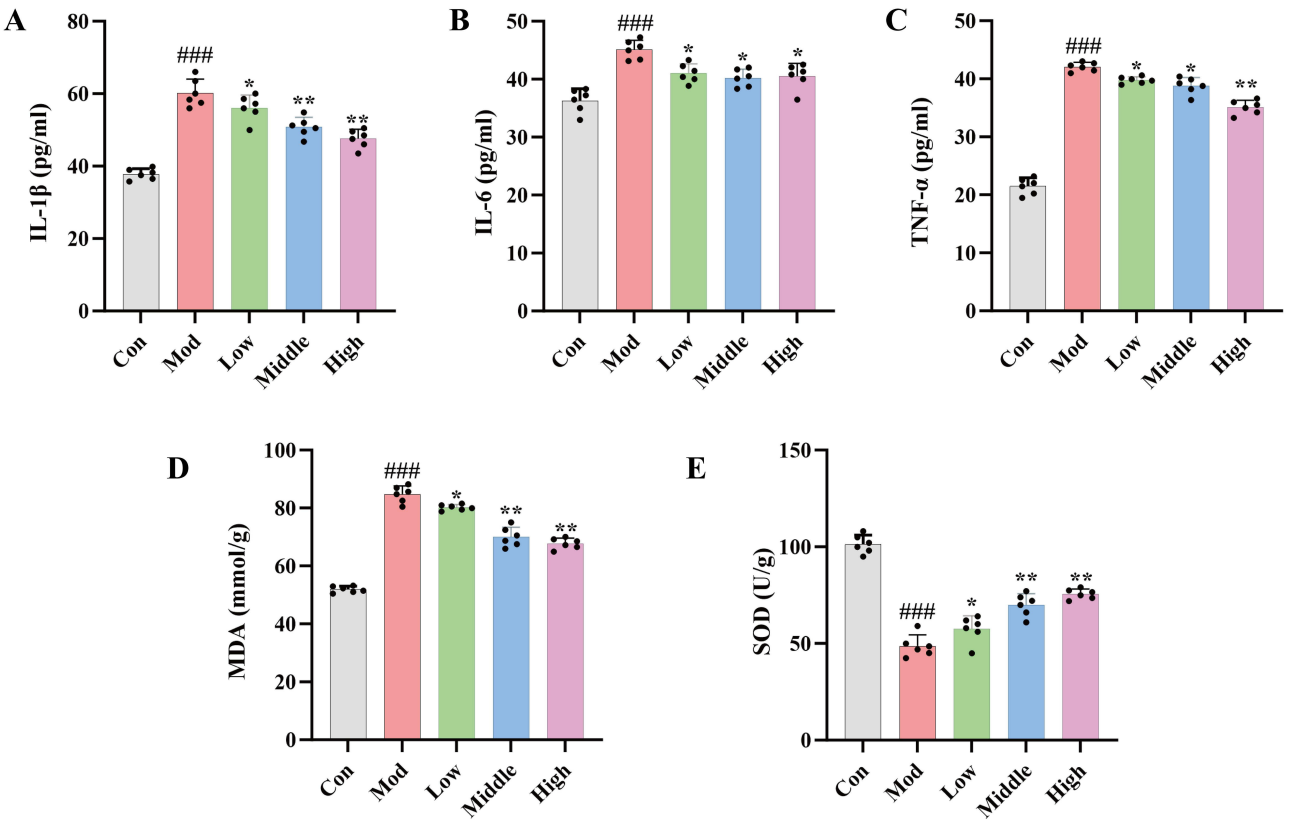


Figure 6 Mouse inflammatory response, oxidative stress indicators. (**A**) Serum levels of the pro-inflammatory factor IL-1β. (**B**) Serum levels of the pro-inflammatory factor IL-6. (**C**) Serum levels of the pro-inflammatory factor TNF-α. (**D**) MDA content in kidney tissues across groups. (**E**) SOD activity in kidney tissues across groups. ###P < 0.0001, compared with the Con group; *P < 0.05, **P < 0.01, compared with the Mod group.

with Lipinski's Rule of Five. Blumenol B 9-O-glucoside, coniferaldehyde, and esculetin ranked highest in network centrality. Molecular docking analysis revealed that esculetin exhibits strong binding ability with the core components of the TGF-β/PI3K-AKT/mTOR pathway, with binding energies below −7 kcal/mol for key renal fibrosis targets including PIK3R1, AKT1, and MTOR. Esculetin, a naturally occurring coumarin, is well-documented to possess anti-inflammatory, antioxidant, and EMT-inhibitory properties. It exerts renal protection by inhibiting the release of pro-inflammatory cytokines, enhancing antioxidant enzymes activity, and preventing EMT.[24–26] In this study, the crude methanol extract of BYJ containing esculetin significantly reduced serum levels of IL-6, IL-1β, TNF-α, and renal tissue MDA levels, while increasing SOD activity and effectively inhibiting renal EMT progression. The pharmacological effects are consistent with the known activities of esculetin, suggesting that esculetin may be a key candidate active component responsible for BYJ's renal protective effects. However, the in vivo experiments in this study utilized the crude methanol extract of BYJ throughout, and no direct verification was performed using pure esculetin. Therefore, it remains unclear whether esculetin is the only active component of BYJ mediating its anti-renal fibrosis effect, and the effective dose and individual efficacy of pure esculetin require further confirmation in subsequent studies.

Network pharmacology analysis revealed a substantial enrichment of BYJ targets in the PI3K-AKT, mTOR, and TGF-β signaling pathways, establishing a clear association between BYJ and the primary mechanisms of renal fibrosis. TGF-β promotes ECM accumulation and myofibroblast activation, positioning it as a crucial regulator of renal fibrosis.[27] Together with TGF-β, PI3K-AKT/mTOR pathway exacerbates fibrosis by modulating renal cell proliferation, metabolic dysregulation, and ECM deposition.[28]

Furthermore, GO enrichment analysis indicated that BYJ significantly influenced 454 molecular functions, 208 cellular components, and 2922 biological processes. Nonetheless, the positive modulation of cell migration, a hallmark

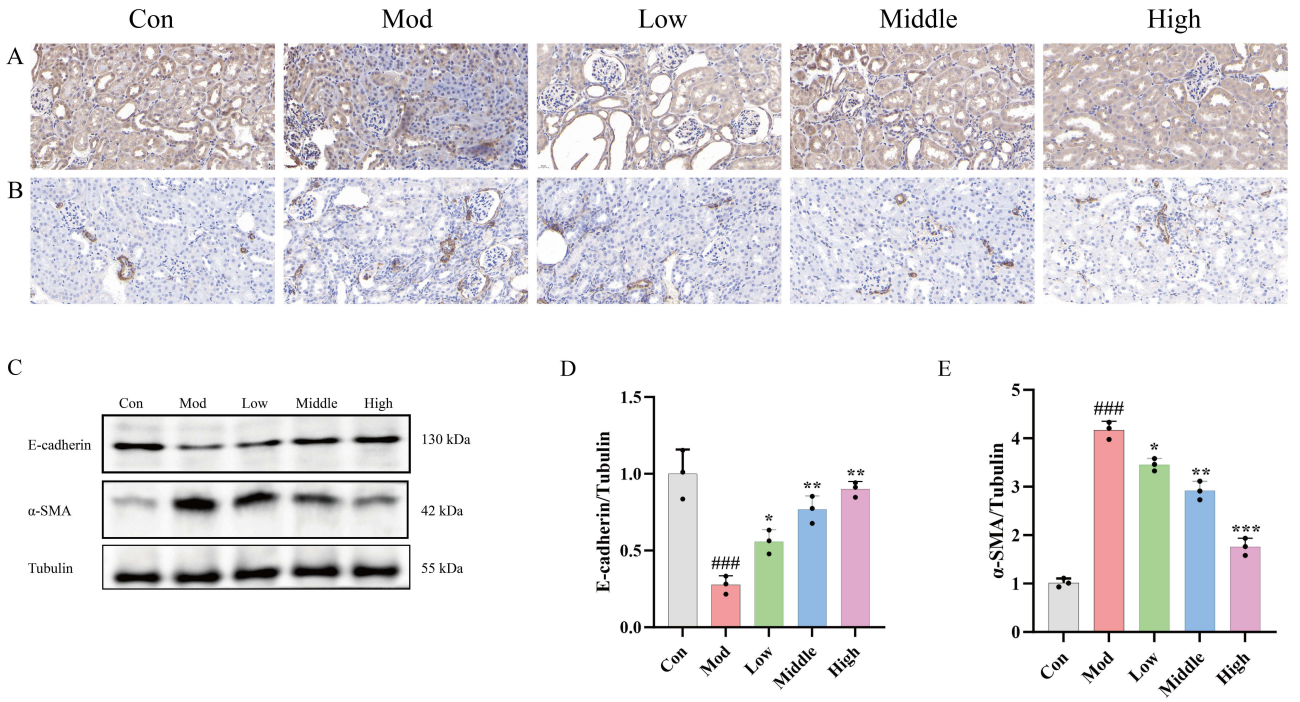


Figure 7 Expression of EMT-related proteins in mice. (**A**) IHC staining of E-cadherin in kidney tissues (magnification, x200). (**B**) IHC staining of α-SMA in kidney tissues (magnification, x200) (**C**) Western blot analysis of E-cadherin and α-SMA in kidney tissues; (**D**) Semi-quantitative analysis of the E-cadherin/Tublin protein ratio; (**E**) Semi-quantitative analysis of the α-SMA/Tublin protein ratio. ###P < 0.0001, compared with the Con group; *P < 0.05, **P < 0.01, ***P < 0.001, compared with the Mod group.

of EMT,[29] suggests that BYJ inhibits EMT by constraining abnormal epithelial cell migration. Significantly, the enhancement of protein phosphorylation directly influences the regulation of PI3K and AKT activity.[30,31] Collectively, these findings demonstrate that the therapeutic effects of the crude BYJ extract in mitigating renal fibrosis are closely associated with the modulation of the TGF-β/PI3K-AKT/mTOR signaling pathway, while other pathways may also be involved and require further investigation.

The TGF-β/PI3K-AKT/mTOR pathway plays a pivotal role in regulating EMT.[32] Upon TGF-β binding to its receptor, PI3K is activated, resulting in AKT phosphorylation and subsequent mTOR activation.[33] This signaling cascade promotes the expression of mesenchymal markers (eg., α-SMA) while inhibiting epithelial markers (eg., E-cadherin), driving epithelial cells towards a fibrogenic, migratory phenotype that accelerates renal fibrosis.[34,] EMT activation in adenine-induced CKD mice was evidenced by increased α-SMA expression, decreased E-cadherin levels, and upregulated expression of TGF-β1, PI3K, p-AKT, and mTOR. BYJ treatment significant attenuated these molecular alterations, suggesting that it modulates the overactivation of the TGF-β/PI3K-AKT/mTOR, thereby suppressing EMT and limiting ECM accumulation.

A limitation of this study is the absence of positive control groups treated with clinical anti-fibrotic and CKD drugs (eg., enalapril, losartan, and pirfenidone), which precludes direct comparison the efficacy of BYJ with that of standard clinical drugs. This study design was driven by the core purpose of clarifying the molecular mechanism underlying BYJ's renal anti-fibrotic effects, rather than comparing its efficacy with other drugs. Simplifying the experimental groups avoids interference from drug interactions during target and pathway identification, and this study first determined the safe and effective dose window of BYJ, laying a foundation for subsequent comparative studies. Despite the lack of direct comparative data, the anti-fibrotic and renal protective effects of BYJ observed in this study are consistent with the therapeutic effects of enalapril and pirfenidone in adenine-induced nephropathy models reported in the literature, which indirectly supports its clinical potential.[12–14] Future studies will include positive control groups, conduct head-to-head efficacy comparisons between BYJ and clinical drugs, explore their potential synergistic effects when combined, and isolate and purify components from BYJ to verify the pharmacological effects of individual components and component combinations.

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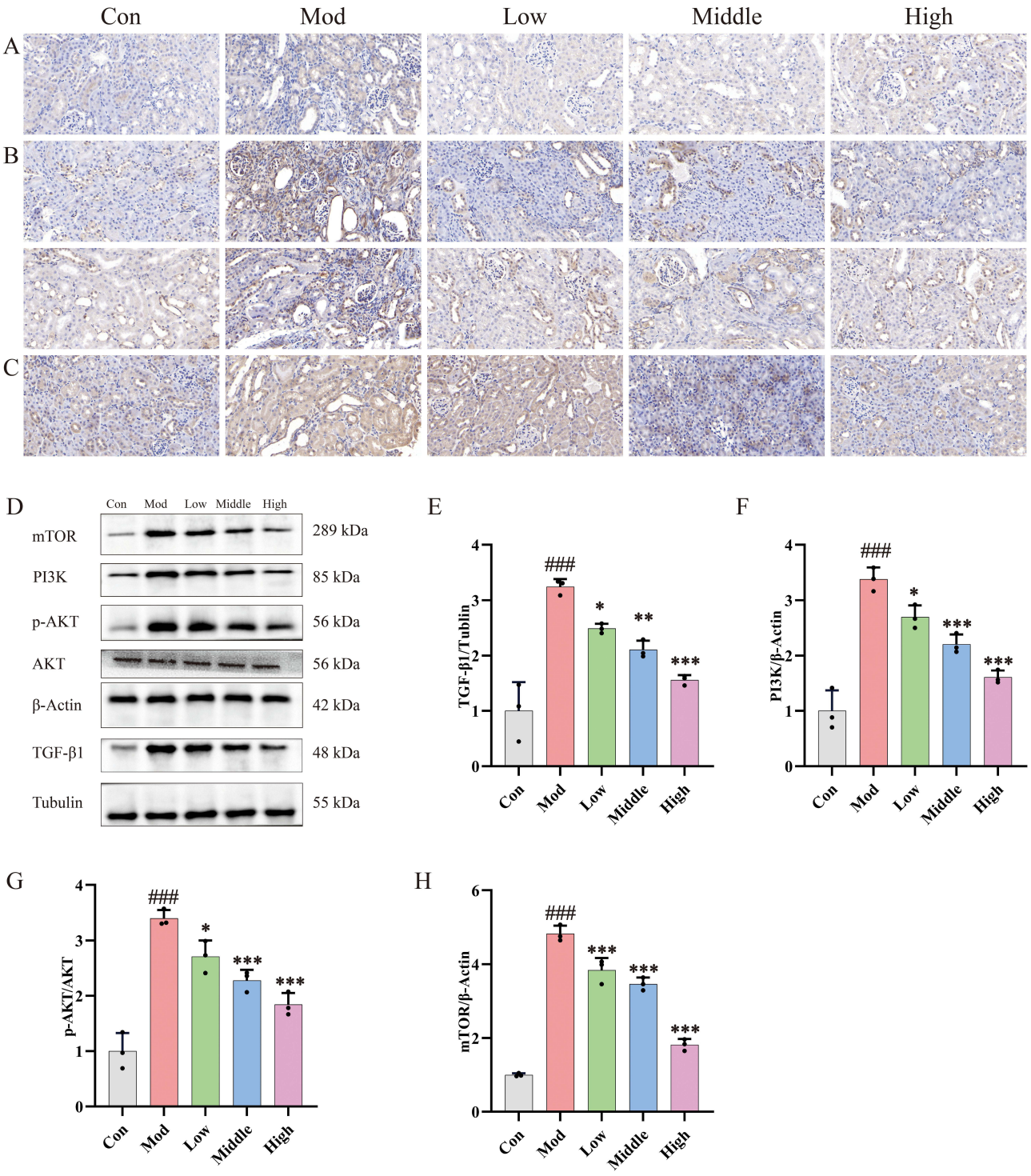


Figure 8 Regulation of the TGF-β/PI3K/AKT/mTOR signaling pathway. (**A**) IHC staining of PI3K in kidney tissues (magnification, x200); (**B**) IHC staining of p-AKT and AKT in kidney tissues (magnification, x200); (**C**) IHC staining of mTOR in kidney tissues (magnification, x200); (**D**) Western blot analysis of TGF-β1, PI3K, p-AKT, AKT and mTOR protein expression in renal tissues, with β-Actin and Tubulin as internal references. (**E**) Semi-quantitative analysis of the TGF-β1/Tublin protein ratio; (**F**) Semi-quantitative analysis of the PI3K/β-actin protein ratio; (**G**) Semi-quantitative analysis of the p-AKT/AKT protein ratios (**H**) Semi-quantitative analysis of the mTOR/β-actin protein ratios. ###P <0.0001, compared with the Con group; *P < 0.05, **P < 0.01, ***P < 0.001.

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Conclusion

In conclusion, this study demonstrates that the methanolic extract of BYJ attenuates the progression of adenine-induced CKD in mice. These protective effects are mediated, at least in part, by the downregulation of the TGF-β/PI3K-AKT/mTOR pathway and the suppression of renal EMT. Although in silico analysis identified esculetin as a putative bioactive constituent, it is important to emphasize that all functional observations herein were derived from the crude herbal extract. Rather than establishing a definitive causal mechanism, our findings establish a strong correlation between BYJ treatment and the modulation of key renal fibrotic signaling cascades. This work lays a robust preclinical foundation for BYJ as a promising therapeutic candidate for CKD and suggests that purified individual components, particularly esculetin, warrant further mechanistic validation in dedicated confirmatory studies.

Data Sharing Statement

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Ethics Approval and Consent to Participate

This study was conducted and reported in accordance with the ARRIVE guidelines (Animal Research: Reporting of In Vivo Experiments). The animal research protocol has been approved by the Ethics Committee of the Yunnan Branch of the Chinese Academy of Medical Sciences Institute of Medicinal Plants (Approval Number: 2025039).

Consent for Publication

Not applicable. This manuscript does not contain any individual person's data in any form (including individual details, images, or videos).

Author Contributions

Jianglong Chen - Conceptualization, Methodology, Writing – original draft, Writing – review and editing. Yunfei Cui - Investigation, Data curation, Formal analysis, Writing – original draft, Rujie Zhou - Investigation, Validation, Visualization, Writing – original draft, Yu Zhu - Resources, Data curation, Software, Writing – original draft, Meijia Chen - Investigation, Formal analysis, Visualization, Writing – original draft, Jinhua Su - Resources, Project administration, Formal analysis, Writing – original draft, Writing – review and editing. Jinhui Wang - Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review and editing. Guang Li- Conceptualization, Supervision, Writing – original draft, Writing – review and editing, Funding acquisition. All authors gave final approval of the version to be published; have agreed on the journal to which the article has been submitted; and agree to be accountable for all aspects of the work.

Funding

This study was jointly funded by the CAMS Innovation Fund for Medical Sciences (2021-I2M-1-031), the Yunnan Provincial Science and Technology Talent and Platform Program (202405AF140073), the Yunnan Provincial Major Science and Technology Special Project (202402AA310041), and the Science and Technology Special Mission for Dai Medicine Industry in Jinghong City, Yunnan Province (202404BI090001).

Disclosure

The authors declare that they have no competing interests.

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